Molecular Modeling in Process Engineering

2025/2026

Project 1

Estimation of quantized properties

The purpose is to determine the following properties as predicted from quantum chemistry theory.

1) Determine the translational zero point energy of a methane molecule in a box having the dimension of 1 m.
2) Determine the energy of the orbitals of atomic hydrogen both from theory and using a quantum chemistry approach.
3) Draw the 1s, 2s, 2px, 2py, and 2pz orbitals of atomic hydrogen on a xy plane.
4) Determine the average distance of the electron in a 1s, 2s, 2px, 2py, and 2pz orbital from the nuclei for atomic hydrogen.